

Reactive Transport Control on Finite-Offset Seismoelectric Interface Responses During Carbonate Dissolution

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Key Points:

- Carbonate dissolution changes hydraulic connectivity and pore fluid conductivity through different pathways
- Conductivity growth can suppress electrokinetic coupling even when permeability increases
- Finite offset waveforms preserve the coefficient level weakening of interface electromagnetic responses

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Abstract

Seismoelectric interface responses can be sensitive to hydraulic and electrical contrasts, but it remains unclear how reactive transport changes these signals as pore networks dissolve. Here we map reactive transport outputs for carbonate dissolution into dynamic electrokinetic properties, Schakel-type interface conversion coefficients, and Liu-type finite-offset waveform synthesis. The simulations indicate that dissolution increases porosity and permeability, but higher Peclet number cases also produce stronger pore-fluid conductivity growth and a marked decrease in the dynamic coupling coefficient. As a result, the interface conversion coefficients and post- T_0 waveform amplitudes decrease most strongly in the cases with the largest permeability increase. These results suggest that seismoelectric monitoring of dissolution should account for the competition between hydraulic connectivity and electrical leakage, rather than interpreting permeability growth alone.

Plain Language Summary

When rock dissolves, pores can become more connected and fluids can move more easily. At the same time, the fluid in the pores can become more electrically conductive. These two changes have opposite effects on seismoelectric signals, which are electrical signals produced when seismic waves pass through water-filled porous rocks. This study uses numerical simulations to test how carbonate dissolution changes these signals. The results show that stronger flow pathways do not necessarily produce stronger seismoelectric responses, because increased fluid conductivity can weaken the electrokinetic source.

1 Introduction

Seismoelectric interface responses provide a way to connect seismic wave propagation with hydraulic and electrical contrasts in porous media. Previous work has shown that these responses can contain information about permeability, saturation, salinity, and layered structure, but most interpretations treat the material state as fixed.

Reactive transport studies show that mineral dissolution can reorganize pore geometry, flow pathways, and pore-fluid chemistry over a much longer time scale. For carbonate media, these changes can increase porosity and permeability while also changing tortuosity and electrical conductivity.

The central challenge is that these changes do not affect the seismoelectric source in the same direction. Permeability growth can strengthen hydraulic communication, whereas conductivity growth can reduce the relative electrokinetic coupling and weaken the interfacial current imbalance.

This study tests how this competition is transferred from reactive transport outputs to dynamic electrokinetic parameters, interface conversion coefficients, and finite-offset waveform amplitudes. We distinguish dissolution time, which describes chemical evolution, from waveform time, which describes the microsecond-scale response of one evolved material state.

2 Model and Analysis Workflow

2.1 Reactive Transport Inputs

We use three reactive transport simulations with $Pe = 0.1, 1,$ and 10 as evolving porous media states. For each time step, porosity, permeability, tortuosity, outlet H^+ concentration, and derived pore-fluid conductivity define the material inputs to the seismoelectric model.

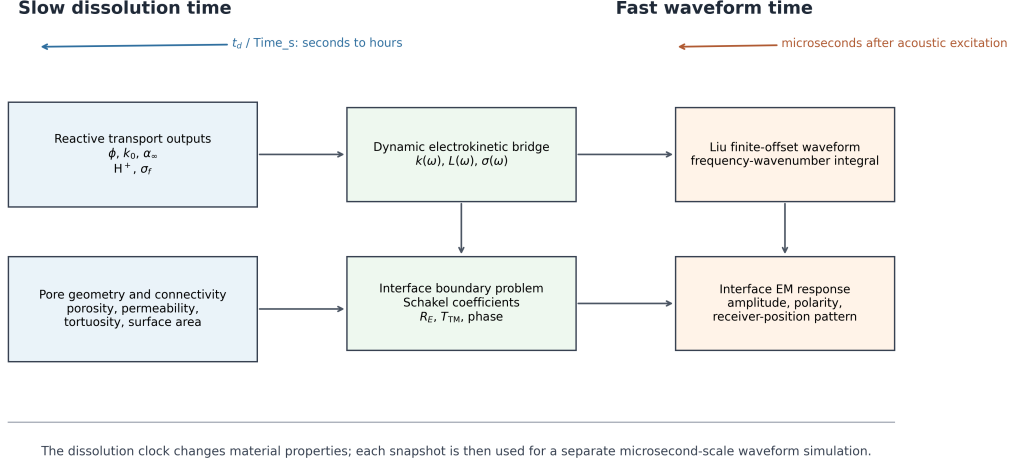


Figure 1. Conceptual workflow and time-scale separation. Reactive transport evolves material properties over dissolution time, whereas the seismoelectric waveform is calculated for each fixed dissolution state over waveform time.

The analysis is restricted to the poroelastic validity window used in the current forward model. This restriction keeps the interpretation focused on the part of dissolution evolution that can be mapped consistently into the Schakel/Pride-type seismoelectric formulation.

2.2 Mapping Reactive Transport Outputs to Dynamic Electrokinetic Properties

The reactive transport variables are converted into frequency-dependent permeability, dynamic electrokinetic coupling $L(\omega)$, and dynamic conductivity $\sigma(\omega)$. This step defines the physical bridge between chemical evolution and the electromagnetic source terms used in the interface problem.

2.3 Interface Conversion Coefficients

For each evolved state, the dynamic properties are inserted into the fluid-porous medium interface boundary value problem. The main coefficient-level quantities are the reflected electric response R_E and transmitted transverse magnetic response T_{TM} .

2.4 Finite-Offset Waveform Synthesis

Finite-offset waveforms are calculated with a frequency-wavenumber spectral integral following the Liu-type VSEP geometry. Each waveform panel is indexed by dissolution time but plotted against waveform time after acoustic excitation.

2.5 Numerical Diagnostics and Response Metrics

The waveform synthesis is checked for pre- T_0 leakage and frequency sampling convergence. Normalized R_E , T_{TM} , $|L|/|\sigma|$, and spectral-ratio metrics are then used as simulation-based indicators of response change.

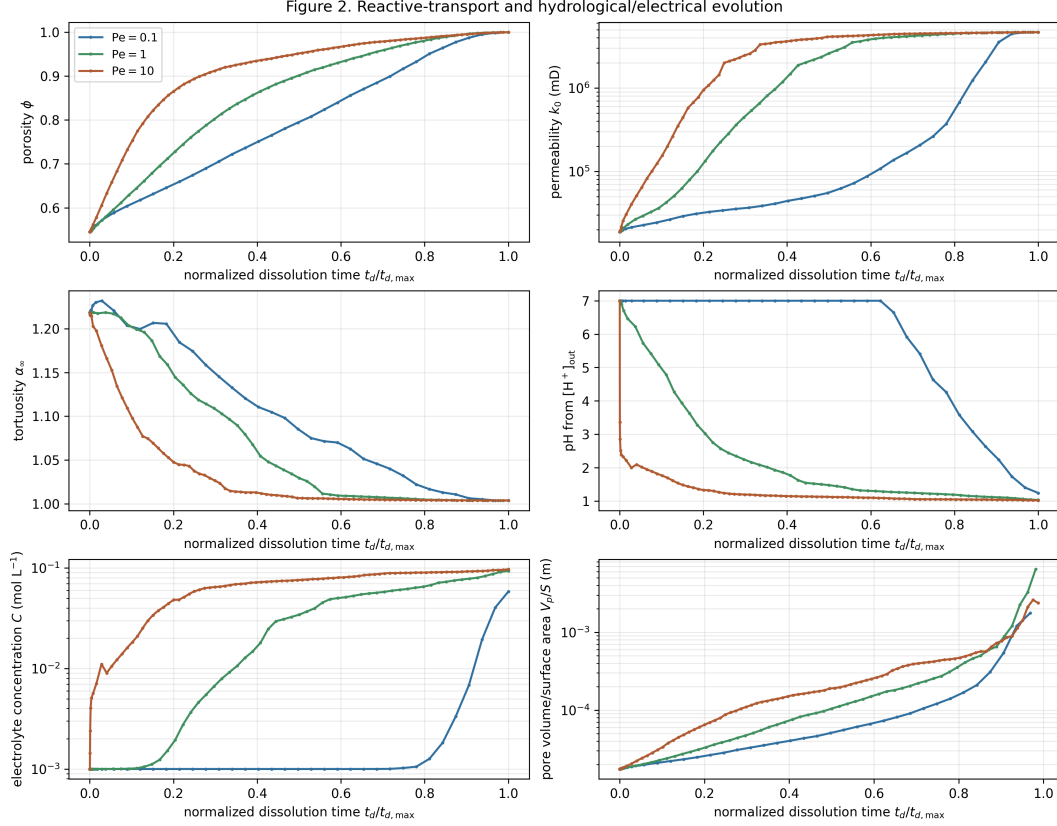


Figure 2. Reactive transport evolution of porosity, permeability, tortuosity, concentration, conductivity, and related state variables. The figure establishes the hydraulic and electrical pathways that feed the seismoelectric model.

3 Results

3.1 Dissolution Changes Hydraulic and Electrical Pathways Together

Dissolution increases porosity and permeability in all Pe cases, but the permeability increase is much larger for $Pe = 1$ and $Pe = 10$ than for $Pe = 0.1$. At the same time, the higher Pe cases develop much stronger pore-fluid conductivity growth, setting up the competition that controls the seismoelectric response. Within the valid window, permeability increases by about 20 times for $Pe = 0.1$ and by more than 200 times for $Pe = 1$ and $Pe = 10$.

3.2 Dynamic Electrokinetic Properties Translate Dissolution Into Source Strength

The dynamic bridge shows that $\sigma(\omega)$ increases with pore-fluid conductivity, whereas $L(\omega)$ decreases strongly in the higher Pe cases. This result indicates that permeability alone is not a sufficient predictor of the interface electromagnetic response. In $Pe = 1$ and $Pe = 10$, $|\sigma|$ increases by about 114–158 times, while $|L|$ decreases to about 1% of its initial value.

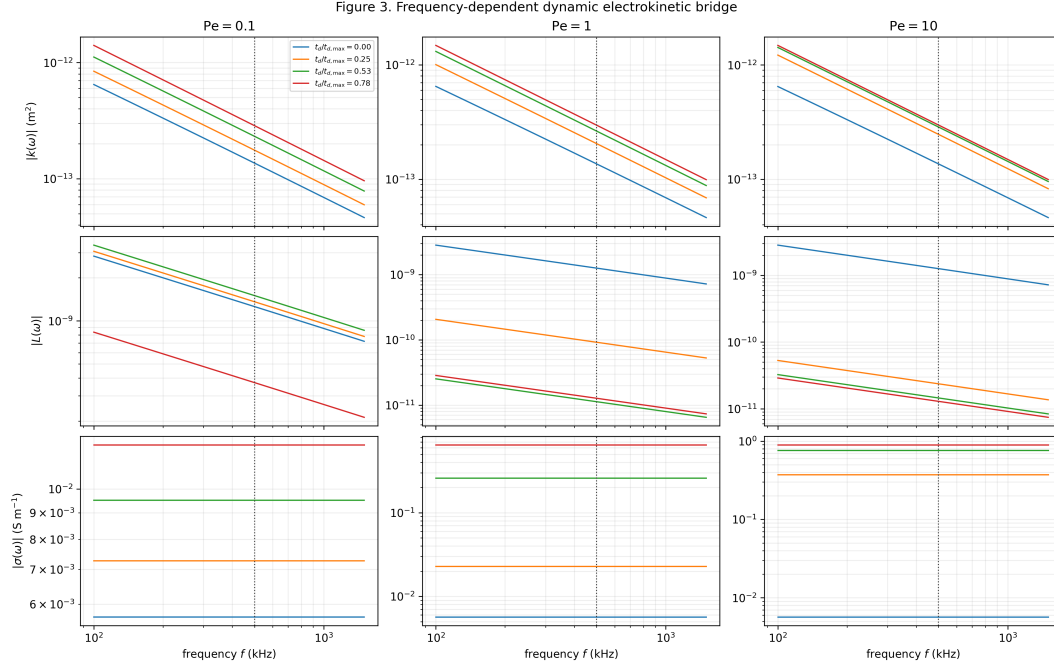


Figure 3. Frequency-dependent dynamic electrokinetic properties derived from the reactive transport states. The opposing trends in $L(\omega)$ and $\sigma(\omega)$ define the main source-strength competition.

3.3 Interface Conversion Coefficients Record Conductivity-Controlled Weakening

The magnitude of R_E decreases moderately for $Pe = 0.1$ but falls by nearly four orders of magnitude for $Pe = 1$ and $Pe = 10$ within the valid window. Frequency-angle heatmaps show that this weakening persists across the sampled incident-wave and frequency space. This contrast indicates that conductivity growth can outweigh the effect of increased hydraulic connectivity in the interface coefficients.

3.4 Finite-Offset Waveforms Preserve the Same Weakening in Waveform Time

The post- T_0 finite-offset response weakens as dissolution proceeds, with the strongest amplitude loss in $Pe = 1$ and $Pe = 10$. The waveform result therefore carries the coefficient-level suppression into the time-domain waveform response in the simulations. Late-stage post- T_0 peaks are about 1.6×10^{15} for $Pe = 0.1$, compared with about 1.8×10^{12} for $Pe = 1$ and 1.3×10^{12} for $Pe = 10$.

3.5 Waveform Diagnostics Support the Post- T_0 Interpretation

Frequency sampling tests show that coarse sampling can create pre- T_0 aliases, but using $n_\omega = 128$ reduces the pre/post- T_0 ratio to about 0.011–0.012 across the waveform panels. This diagnostic supports the interpretation that the plotted main response is not dominated by pre-arrival numerical leakage.

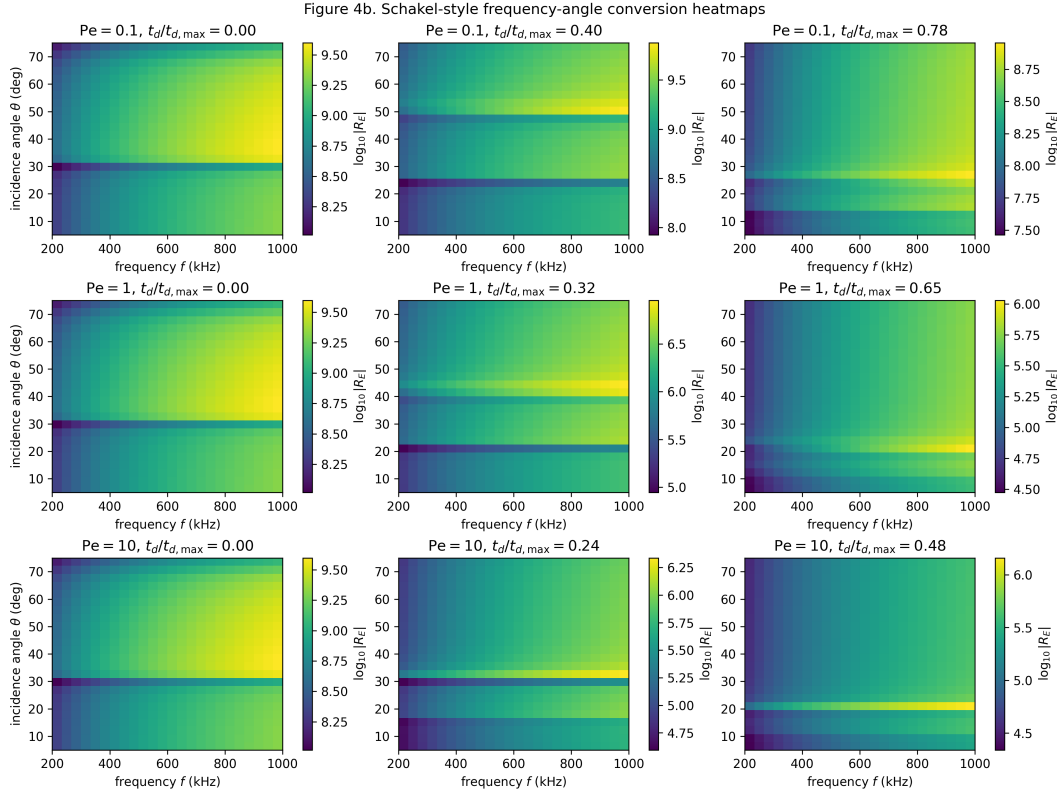
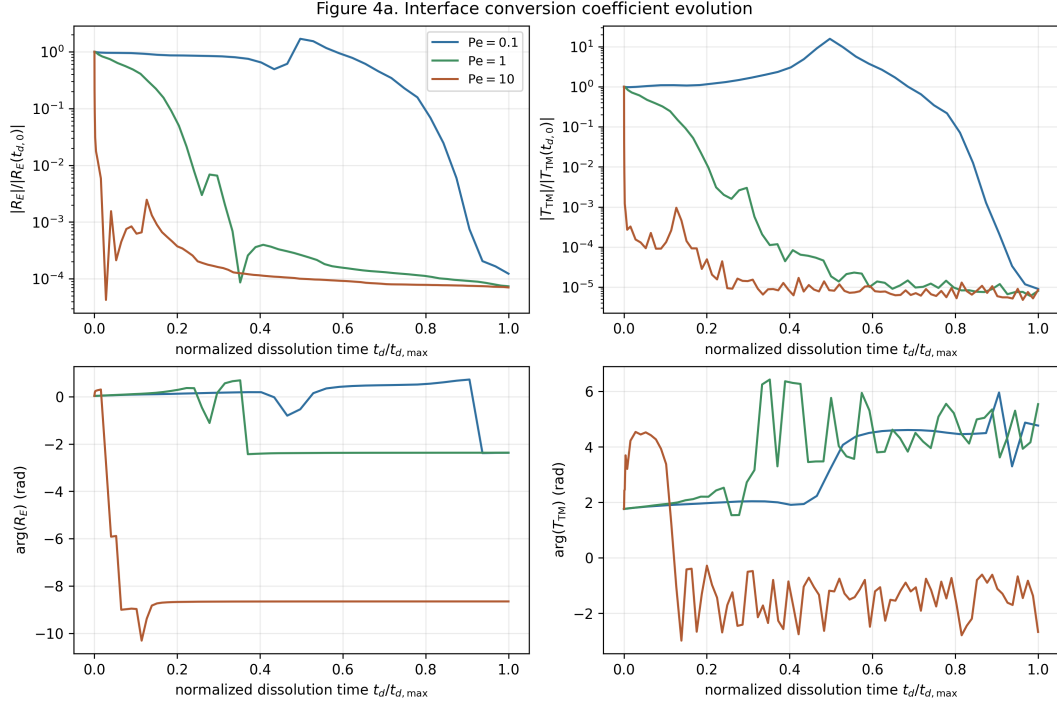


Figure 4. Interface conversion coefficients during dissolution. The time series show how dynamic property changes are transferred into conversion strength, and the frequency-angle heatmaps separate wavefield dependence from the dissolution trend.

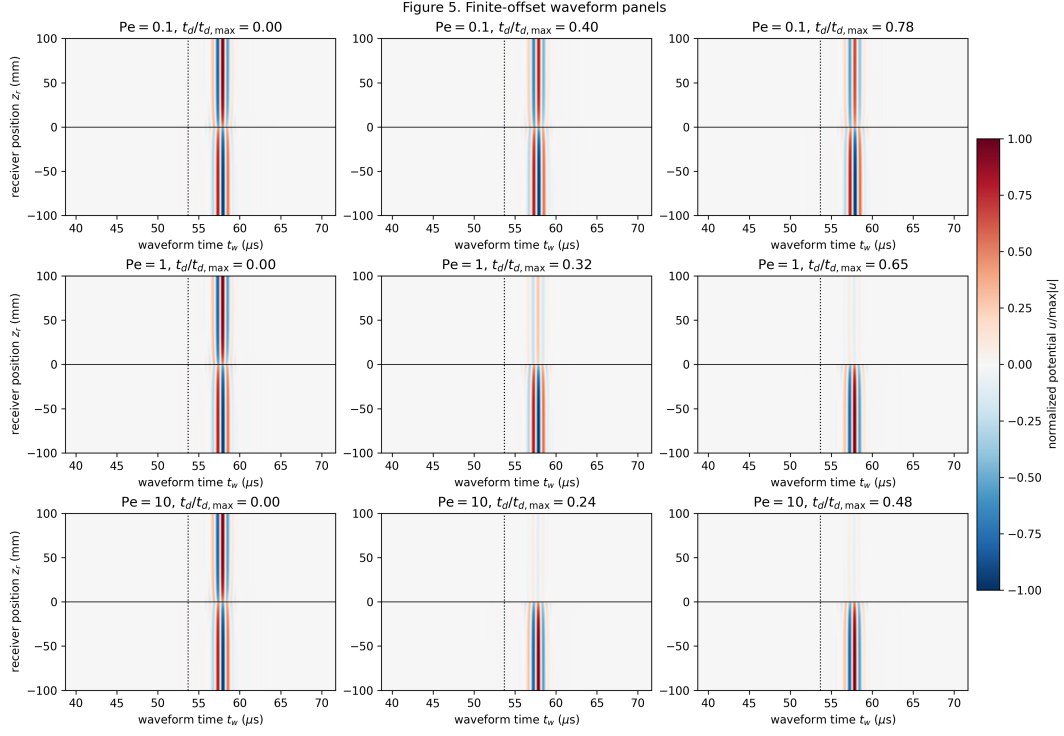


Figure 5. Finite-offset waveform panels for selected dissolution states. Dissolution time indexes the material state, while waveform time records the response after acoustic excitation.

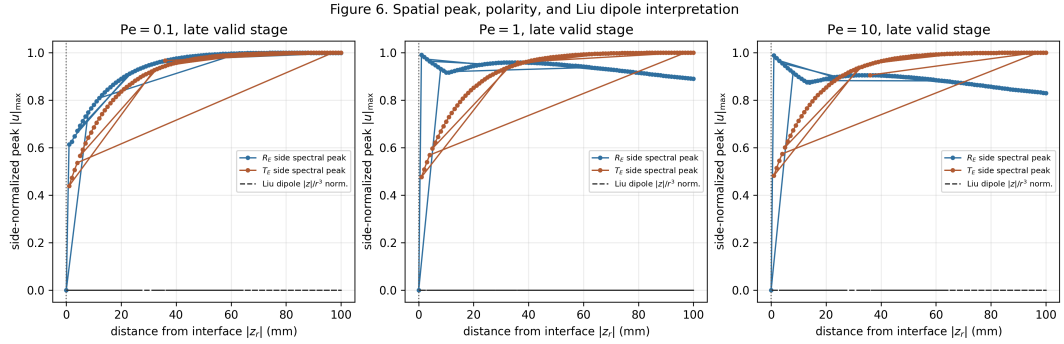


Figure 6. Spatial peak distribution, polarity behavior, and normalized dipole interpretation. The figure tests whether the modeled response has the expected finite-offset spatial organization.

3.6 Spatial Peak Patterns Are Consistent With an Interface Source Geometry

The finite-offset peak amplitudes vary with receiver position and show polarity differences across the interface. The Liu-type dipole pattern is used only as a normalized spatial interpretation, not as an additional multiplier on the forward waveform. This subsection functions as a geometry consistency check before the mechanism decomposition.

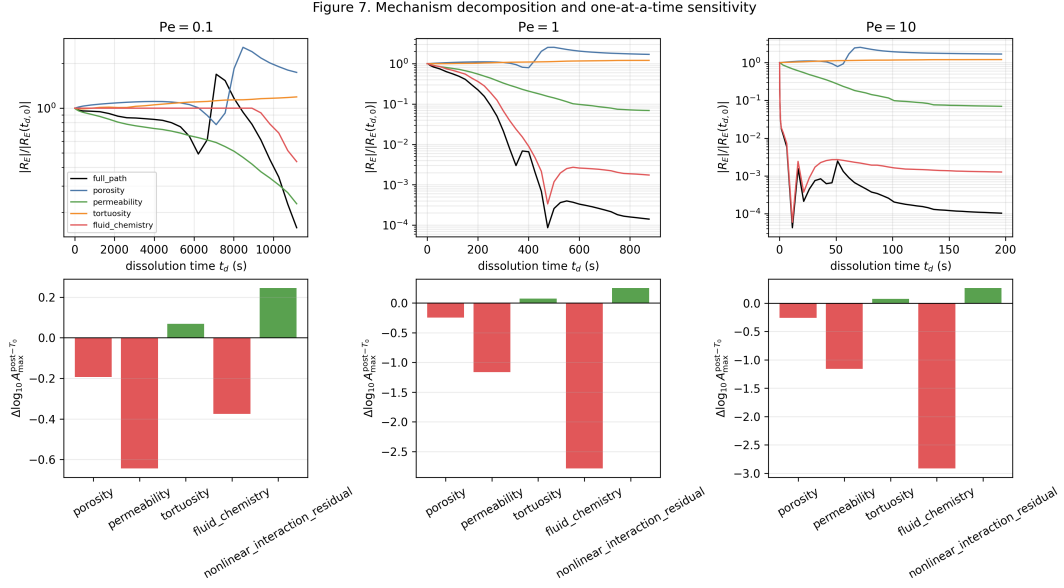


Figure 7. Mechanism decomposition using one-at-a-time parameter substitutions. The figure separates the contributions from porosity, permeability, tortuosity, and fluid chemistry pathways.

3.7 Mechanism Decomposition Identifies the Dominant Parameter Pathway

One-at-a-time substitutions show that the largest negative contribution in $Pe = 1$ and $Pe = 10$ comes from the fluid chemistry pathway. The late-stage amplitude loss is therefore interpreted mainly as conductivity-related weakening of electrokinetic coupling, with permeability and nonlinear interactions acting as secondary controls in the tested configuration. For $Pe = 1$ and $Pe = 10$, the fluid chemistry pathway contributes $\log_{10}$ peak changes of about -2.78 and -2.91, respectively.

3.8 Normalized Metrics Provide Simulation-Based Monitoring Indicators

Normalized R_E and $|L|/|\sigma|$ decrease more strongly than the simple high-over-low frequency ratio in the higher Pe cases. These metrics are treated as diagnostic quantities from the simulations, not as evidence of field detectability. The normalized R_E metric decreases to about 0.16 for $Pe = 0.1$ and to about 10^{-4} for $Pe = 1$ and $Pe = 10$.

4 Discussion

4.1 Why Stronger Hydraulic Connectivity Can Produce a Weaker Interface Response

The main result is not a monotonic relation between dissolution and seismoelectric amplitude. Instead, dissolution can increase hydraulic connectivity while increasing electrical leakage, and the latter effect can dominate the interface electromagnetic source under the simulated conditions.

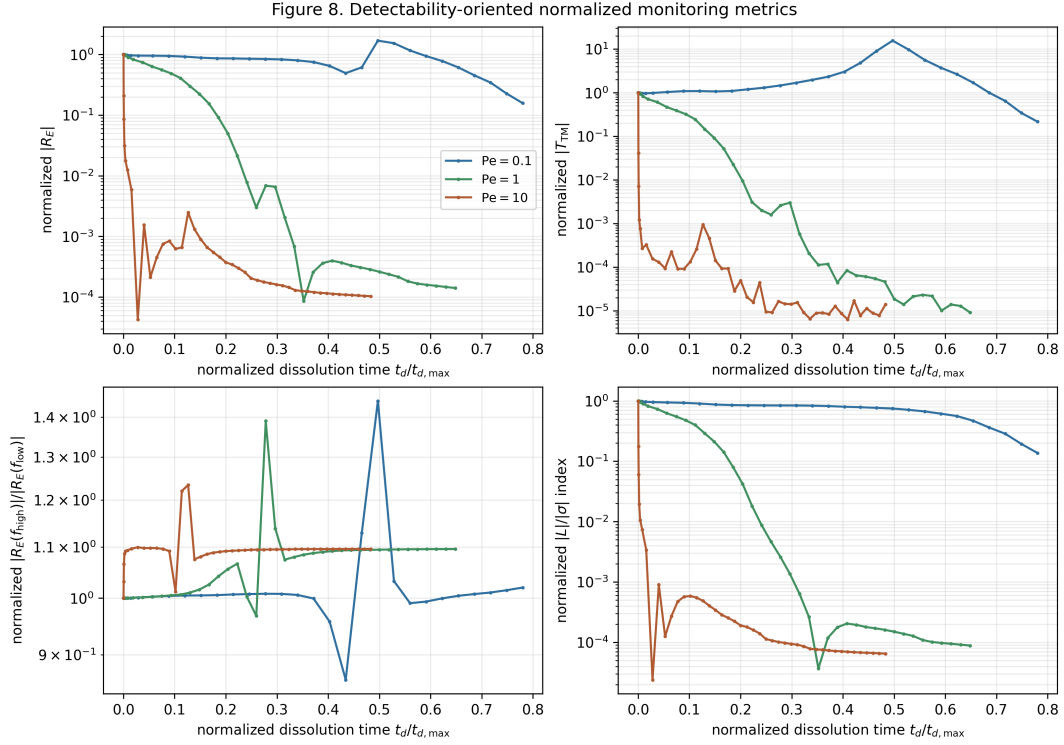


Figure 8. Normalized response metrics during dissolution. The figure compares coefficient-based and spectral-ratio indicators of response change.

4.2 Role of Peclet Number in the Coupled Response

The $Pe = 1$ and $Pe = 10$ cases develop much larger permeability and conductivity changes than $Pe = 0.1$. Their weaker interface responses indicate that transport regime affects the seismoelectric signal through coupled hydraulic and chemical pathways, not through permeability alone.

4.3 Implications for Seismoelectric Monitoring of Reactive Systems

The results suggest that seismoelectric monitoring of dissolution should track both hydraulic and electrical state variables. Amplitude loss during dissolution should not be interpreted directly as reduced permeability without considering pore-fluid conductivity and coupling strength.

4.4 Limitations and Future Work

The present workflow uses simulated reactive transport outputs and a forward seismoelectric model, so the conclusions are limited to the tested geometry, parameter mapping, and poroelastic validity window. Future work should separate independent fluid conductivity and ζ -potential effects and compare the predicted trends with laboratory or field measurements.

5 Conclusions

This study links reactive transport evolution to finite-offset seismoelectric interface responses through dynamic electrokinetic properties and interface conversion coef-

ficients. Dissolution increases porosity and permeability, but higher Pe cases also increase pore-fluid conductivity enough to suppress $L(\omega)$, R_E , T_{TM} , and post- T_0 waveform amplitudes.

The mechanism decomposition indicates that fluid chemistry is the main negative pathway in the strongest amplitude losses. The practical implication is that seismoelectric response changes during dissolution should be interpreted as the outcome of competing hydraulic and electrical effects.

Open Research Section

The reactive transport input tables, seismoelectric modeling scripts, figure-generation scripts, and processed waveform diagnostics will be archived in a public repository before submission. The present manuscript outline uses local files under `paper_figure_sequence_analysis`; the final paper should replace this sentence with the repository DOI.

As Applicable – Inclusion in Global Research Statement

This numerical modeling study does not currently involve field sampling, permits, or local community data. The final manuscript should update this statement if new collaborators, field data, or institutional requirements are added.

Conflict of Interest declaration

The authors declare that they have no conflicts of interest relevant to this manuscript.

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